

1. Definition

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A Composite Key is a key formed by two or more columns together that uniquely identify a row in a table.

None of the individual columns can uniquely identify the row on their own, but their combination can.



Neither column is sufficient alone to represent enrollement. Many-to-Many Relationships

Students ↔ Courses

One student can take many courses.

One course can have many students.

Enrollment table

StudentID	CourseID
101	DBMS
101	OS

↑ Composite Key ↑

(StudentID, CourseID) , **Properties** : uniqueness , minimal , multiple columns , A Composite Key can be selected as the Primary Key.

Disadvantage 1 — Larger Indexes

Indexes store multiple columns.
More storage is required.

Disadvantage 2 — Slower Joins

Joining tables on multiple columns is generally more expensive than joining on a single integer ID.

Disadvantage 3 — Larger Foreign Keys

If another table references this Composite Key,
it must store all key columns.

Example

Instead of
EnrollmentID

another table must store

StudentID

CourseID

Creating a Composite Primary Key

```
CREATE TABLE Enrollment
(
    StudentID INT,
    CourseID INT,

    PRIMARY KEY(StudentID, CourseID)
);
```

Composite Foreign Key

Suppose another table stores marks.

```
CREATE TABLE Marks
(
    StudentID INT,
    CourseID INT,
    Marks INT,

    FOREIGN KEY(StudentID, CourseID)
    REFERENCES Enrollment(StudentID, CourseID)
);
```

Notice that the Foreign Key must reference both columns together.